

Merton Jump-Diffusion: Jump Risk Premium

Week 7 — Implied Volatility Surface and Model Limitations

Computational Finance — Group 3

1 Introduction

This report analyses the performance of the Merton (1976) jump-diffusion model against QQQ market option prices through the lens of the implied volatility surface. We use the calibrated parameters from Week 6 to construct model implied volatility (IV) surfaces, compare them with market surfaces, and diagnose the fundamental structural limitations of the model.

The calibrated parameters used throughout (regularised, $\alpha = 0.01$) are:

$$\sigma = 0.1741, \quad \lambda = 0.4860, \quad \mu_J = -0.1967, \quad \sigma_J = 0.2161.$$

These represent a moderate-jump regime pulled away from the boundary optimum $\mu_J = -0.50$ by L1 regularisation, producing more economically interpretable parameters while incurring only a small increase in RMSE ($1.40\% \rightarrow 1.50\%$).

2 Implied Volatility Surfaces

2.1 Construction

For each strike K and maturity T in the calibration dataset, the Merton model price is computed via the Poisson-weighted Black–Scholes sum:

$$C^{\text{Merton}}(K, T) = \sum_{n=0}^{\infty} \frac{e^{-\lambda'T} (\lambda'T)^n}{n!} C^{\text{BS}}(S_0, K, T, r_n, \sigma_n), \quad (1)$$

where $\lambda' = \lambda(1 + \kappa)$, $r_n = r - \lambda\kappa + n\mu_J/T + n\sigma_J^2/(2T)$, $\sigma_n = \sqrt{\sigma^2 + n\sigma_J^2/T}$, and $\kappa = e^{\mu_J + \sigma_J^2/2} - 1$.

The implied volatility is recovered by inverting the Black–Scholes formula numerically via Brent’s method. OTM puts are used for $K < S_0$ and OTM calls for $K > S_0$ to minimise bid–ask spread effects.

2.2 Results

The three surfaces — market, Merton model, and residual (market minus model) — are computed on a grid of 45 strikes spanning the calibration range (\$626–\$845) and all six available maturities ($T = 0.077$ to 1.052 yr).

The residual surface reveals two systematic failure regions:

1. **Deep OTM puts** ($K/S_0 < 0.92$): market IV exceeds model IV by up to 3–5 vol points, reflecting the market’s demand for crash protection that the lognormal jump distribution underweights.
2. **Short maturities** ($T < 0.15$ yr): the model smile is too flat relative to the steep market skew observed at 1–2 month horizons.

3 Smile Decay: Analytical Proof and Empirical Evidence

3.1 Analytical Result

Proposition 1 (Merton smile decay). *Under the Merton jump-diffusion model, the implied volatility skew defined as $\text{Skew}(T) = \sigma_{\text{IV}}(0.90 S_0, T) - \sigma_{\text{IV}}(1.10 S_0, T)$ decays asymptotically as*

$$\text{Skew}(T) \sim \frac{A}{\sqrt{T}}, \quad T \rightarrow \infty,$$

for some constant $A > 0$ depending on $(\lambda, \mu_J, \sigma_J)$.

Proof. The characteristic function of $\log S_T$ under Merton is

$$\varphi_T(u) = \exp \left[T \left(iu\tilde{r} - \frac{1}{2}\sigma^2 u^2 + \lambda \left(e^{iu\mu_J - \frac{1}{2}\sigma_J^2 u^2} - 1 \right) \right) \right],$$

where $\tilde{r} = r - \lambda\kappa$. For large T , the central limit theorem for compound Poisson processes gives

$$\frac{\log S_T - \mu_T T}{\sqrt{V_T T}} \xrightarrow{d} \mathcal{N}(0, 1),$$

where $V_T = \sigma^2 + \lambda(\mu_J^2 + \sigma_J^2)$ is the total variance per unit time. The third cumulant of $\log S_T$ is $\kappa_3(T) = \lambda T \mu_J(\mu_J^2 + 3\sigma_J^2)$, growing linearly in T , while the standard deviation grows as $\sqrt{T}$. The skewness (normalised third cumulant) is therefore

$$\gamma_1(T) = \frac{\kappa_3(T)}{(\text{Var}[\log S_T])^{3/2}} = \frac{\lambda \mu_J(\mu_J^2 + 3\sigma_J^2)}{V_T^{3/2}} \cdot \frac{1}{\sqrt{T}}.$$

Since the IV skew is proportional to the return skewness to leading order (Carr–Wu approximation), we obtain $\text{Skew}(T) \sim A/\sqrt{T}$. $\square$

Remark 1. *The market implied volatility skew decays empirically as $T^{-0.086}$ (estimated from QQQ data), far slower than the theoretical $T^{-0.5}$ of the Merton model. This is the fundamental limitation: the model cannot simultaneously fit both the level and the term structure of the skew.*

3.2 Empirical Evidence

The fitted decay rates from QQQ data are:

	Slope	Interpretation
Merton model	−0.500	Theoretical $O(1/\sqrt{T})$
QQQ market	−0.086	Near-flat; persistent skew

Table 1: Smile decay exponents: Merton vs QQQ market.

The near-flat market skew term structure is consistent with the presence of stochastic volatility. Under the Heston model the skew decays as $O(T^{-1/2})$ for short T but flattens toward a constant for long T , producing an intermediate effective exponent compatible with the observed -0.086 .

4 Short-Maturity Smile Analysis ($T \approx 1\text{M}$)

4.1 Where the Merton Model Fits Well

At the shortest available maturity ($T = 0.077\text{ yr}$, approximately 1 month, 424 contracts across 109 strikes), the model performs acceptably in:

- **Near-the-money** ($0.95 \leq K/S_0 \leq 1.05$): mean absolute residual of 0.32% in IV, consistent with bid–ask spread noise.
- **Smile convexity**: the model correctly produces a positive butterfly (smile shape) rather than a flat line.
- **Overall ATM level**: matched to within 0.5%.

4.2 Where the Merton Model Fails

1. **Deep OTM puts** ($K/S_0 < 0.92$): model undershoots market IV by 3–6 vol points. The lognormal jump distribution assigns insufficient probability mass to large negative moves.
2. **Far OTM calls** ($K/S_0 > 1.08$): slight overshoot, since negative μ_J skews the jump distribution asymmetrically downward.
3. **Left-wing slope**: the market put skew is steeper than the model can produce even at the boundary optimum $\mu_J = -0.50$. This is a structural, not parametric, limitation.

The fit improves monotonically with maturity. At $T > 0.5\text{ yr}$ the model captures the smile shape well; at $T < 0.2\text{ yr}$ the left-wing failure is most pronounced.

5 Calibration Performance by Maturity

Table 2: Calibration RMSE by expiry date (Merton vs Kou).

Expiry	T (yr)	N	Merton RMSE	Kou RMSE	Difference
2026-06-26	0.077	109	1.341%	1.363%	−0.022%
2026-07-31	0.173	139	1.493%	1.473%	+0.021%
2026-08-21	0.230	44	0.732%	0.720%	+0.012%
2026-12-18	0.556	44	1.153%	1.155%	−0.002%
2027-03-19	0.805	44	1.611%	1.616%	−0.005%
2027-06-17	1.052	44	1.708%	1.709%	−0.001%
Overall		424	1.400%	1.398%	−0.002%

The Kou model’s improvement over Merton is negligible (0.002% overall), suggesting that the distributional form of the jump size is not the primary source of model error. The dominant limitation is the absence of stochastic volatility dynamics.

6 Model Comparison and Richer Alternatives

6.1 Summary Table

Table 3: Comparison of option pricing models and how they address Merton’s limitations.

Model	Jump type	Params	Smile decay	Key advantage
Black–Scholes	None	1	Flat (none)	Baseline only
Merton (unreg.)	Normal	4	$O(T^{-0.5})$ too fast	Simple, closed form
Merton (reg.)	Normal	4	$O(T^{-0.5})$ too fast	Stable parameters
Kou	Asymm. exp.	5	$O(T^{-0.5})$ too fast	Heavier tails
Heston	Stoch. vol	5	Slow ($\sim T^{-0.3}$)	Persistent skew
Bates	Normal + stoch. vol	9	Slow, matches market	All main failures
Inf. activity Levy	∞ jumps	3–4	Correct power law	Excellent $T \rightarrow 0$

6.2 Addressing Each Failure

Failure 1: Smile decays too fast ($O(1/\sqrt{T})$). The Heston (1993) stochastic volatility model introduces mean-reverting variance

$$dv_t = \kappa(\bar{v} - v_t) dt + \xi\sqrt{v_t} dW_t^v,$$

which generates a persistent skew whose term structure is controlled by the mean-reversion speed κ . For small κ (slow mean reversion) the model reproduces the near-flat market skew term structure.

Failure 2: Deep OTM put wing underestimated. The Kou (2002) model replaces the Gaussian jump distribution with an asymmetric double-exponential:

$$f_J(x) = p\eta_1 e^{-\eta_1 x} \mathbf{1}_{x>0} + (1-p)\eta_2 e^{\eta_2 x} \mathbf{1}_{x<0}.$$

The calibrated parameters ($\eta_2 = 2.44$, heavy downside tail) slightly improve the OTM put wing. However, the improvement is marginal because the fundamental issue is the absence of stochastic volatility, not the jump distribution.

Failure 3: Short-maturity behaviour ($T \rightarrow 0$). Infinite activity Levy processes (Variance Gamma, Normal Inverse Gaussian, CGMY) allow infinitely many small jumps per unit time. These generate a steep short-maturity smile decaying with the correct empirical power law, because the $T \rightarrow 0$ smile curvature is determined by the Blumenthal–Gettoor index of the process rather than a central-limit-theorem argument.

Industry solution: Bates model. The Bates (1996) model combines Heston stochastic volatility with Merton normal jumps:

$$dS_t = (r - \lambda\kappa) S_t dt + \sqrt{v_t} S_t dW_t^S + (e^J - 1) S_t dN_t, \quad dv_t = \kappa(\bar{v} - v_t) dt + \xi\sqrt{v_t} dW_t^v.$$

It addresses all three failures: stochastic volatility provides slow skew decay and term structure, while jumps sharpen the short-maturity smile and generate the asymmetric wing.

7 Empirical vs Risk-Neutral Jump Parameters

The calibrated risk-neutral intensity $\lambda^Q = 0.163$ jumps/year is substantially lower than the physical (BNS-estimated) intensity $\lambda^P \approx 34.1$ (single-condition) or $\lambda^P \approx 1.38$ (two-condition strict filter). The *jump risk premium* is:

$$\text{JRP} = \lambda^Q \cdot \mathbb{E}^Q[J^2] - \lambda^P \cdot \mathbb{E}^P[J^2].$$

Table 4: Jump risk premium decomposition.

Component	Single-condition	Two-condition
λ^Q	0.163	0.163
$\mathbb{E}^Q[J^2]$	0.357	0.357
Jump variance (risk-neutral)	0.058	0.058
λ^P	34.082	1.377
$\mathbb{E}^P[J^2]$	0.000374	0.003961
Jump variance (physical)	0.01274	0.005454
JRP	4.575	10.688

The large positive JRP indicates that investors pay a significant premium for jump insurance: the risk-neutral measure amplifies jump variance relative to the physical measure, consistent with expensive OTM put prices relative to historical jump frequencies.

8 Conclusion

This analysis identifies three structural limitations of the Merton jump-diffusion model when applied to QQQ options:

1. **Smile decays too fast:** the theoretical $O(1/\sqrt{T})$ decay is six times faster than the empirical market rate of $T^{-0.086}$. This is an intrinsic consequence of the compound Poisson structure and cannot be remedied by re-calibration.
2. **Deep OTM put wing underestimated:** the Gaussian jump distribution assigns insufficient probability to extreme negative returns, producing a smile that is too flat in the left wing.
3. **Short-maturity smile too shallow:** at $T < 0.15$ yr the model underpredicts the steep market skew, particularly where infinite-activity Levy behaviour dominates.

The Bates model (Heston + Merton jumps) addresses all three failures and represents the natural extension. For applications where short-maturity precision is critical, infinite activity Levy processes (VG, NIG, CGMY) provide better $T \rightarrow 0$ behaviour at lower parameter cost.

Nonetheless, the Merton model achieves a calibration RMSE of 1.40% across 424 QQQ contracts — comparable to the more complex Kou model (1.40%) — making it a competitive choice when interpretability and computational tractability are prioritised over marginal accuracy gains.